

Fiber accelerometer — Solution

Y26 (2026) — English translation

A1

0.20 pts

Express the change in diameter of the winding region δD in terms of D , h and δh .

Let's express the volume of rubber:

$$V = \pi D^2 h$$

Let's differentiate the expression:

$$2D \delta D h + D^2 \delta h = 0$$

From here:

Answer:

$$\delta D = -\frac{D}{2h} \delta h$$

A2

0.10 pts

The change in total length of wire δL is proportional to δh

$$\delta L = \Lambda \delta h.$$

Express Λ in terms of N , D , h . The number of revolutions N is strictly fixed.

$$\delta L = \pi N \delta D = -\frac{\pi N D}{2h} \delta h$$

Answer:

$$\Lambda = -\frac{\pi N D}{2h}$$

A3

0.60 pts

Find the strain energy of the wire W . Express your answer using N , D , h , d , E and δh

The volumetric strain energy density is expressed as:

$$w = \frac{E \delta L^2}{2L^2}$$

Wire volume:

$$V = \frac{\pi^2 d^2 N D}{4}$$

Let us express the answer in terms of the quantities specified in the condition:

Answer:

$$W = \frac{E \pi^2 N D d^2 \delta h^2}{32 h^2}$$

A4

0.50 pts

Let at some moment of time the body is displaced ($y \neq 0$) and the load is not in equilibrium ($x \neq y$). Obtain the equation of motion of the load as

$$\ddot{x} + \omega_0^2 x = Ay.$$

Express ω_0 and A in terms of M, h, N, D, d and E .

The total energy is equal to:

$$2W = \frac{E\pi^2 N D d^2 (x - y)^2}{16h^2}$$

Let us express the force acting on the load:

$$F = \frac{\partial(2W)}{\partial(\delta h)} = M\ddot{x}$$

The final equation:

$$\ddot{x} + \frac{E\pi^2 N D d^2 x}{8Mh^2} = \frac{E\pi^2 N D d^2 y}{8Mh^2}$$

From here:

Answer:

$$\omega_0 = \sqrt{\frac{E\pi^2 N D d^2}{8Mh^2}}$$

$$A = \frac{E\pi^2 N D d^2}{8Mh^2}$$

A5

0.50 pts

Express L_0 in terms of $y_0, \omega/\omega_0, \Lambda$ and ζ .

Let's use the method of complex amplitudes:

$$x_0(-\omega^2 + 2\zeta i\omega\omega_0 + \omega_0^2) = \omega_0^2 y_0$$

We express $h_0 = x_0 - y_0$:

$$h_0 = y_0 \left(-1 + \frac{\omega_0^2}{\omega_0^2 - \omega^2 + 2\zeta i\omega\omega_0} \right)$$

Taking into account that $L_0 = \Lambda h_0$, we write the answer:

Answer:

$$L_0 = \Lambda y_0 \left(-1 + \frac{1}{1 - \frac{\omega^2}{\omega_0^2} + 2i\zeta \frac{\omega}{\omega_0}} \right) = \Lambda y_0 \frac{-\frac{\omega^2}{\omega_0^2} + 2i\zeta \frac{\omega}{\omega_0}}{1 - \frac{\omega^2}{\omega_0^2} + 2i\zeta \frac{\omega}{\omega_0}}$$

A6

1.00 pts

Indicate the correspondence between frequency ranges and sensor operating modes. Express K_s and K_a in terms of M, h, N, D, d and E .

Let us consider the behavior of the load in the low-frequency limit. Let us consider one harmonic of oscillations y . Let us simplify the expression for L_0 in the limit $\zeta\omega_0 \ll \omega \ll \omega_0$:

$$L_0 \approx \frac{\Lambda\omega^2 y_0}{\omega_0^2},$$

which corresponds to $L = -\frac{\Lambda\ddot{y}}{\omega_0^2}$. When adding several low-frequency harmonics, this relationship still holds. Let us express K_a :

$$K_a = -\frac{\omega_0^2}{\Lambda} = \frac{E\pi d^2}{4Mh}$$

Now consider the behavior of the load in the high-frequency limit. Let us similarly consider one harmonic of oscillations y . Let us simplify the expression for L_0 in the limit $\omega \gg \omega_0$:

$$L_0 \approx -\Lambda y_0$$

When adding several high-frequency harmonics, this relation is still satisfied. In this case $K_s = -\frac{1}{\Lambda} = \frac{2h}{N\pi D}$.

Answer: With $\zeta\omega_0 \ll \omega \ll \omega_0$ the sensor works as an accelerometer, and with $\omega \gg \omega_0$ – as a seismograph.

$$K_s = \frac{2h}{N\pi D}$$

$$K_a = \frac{E\pi d^2}{4Mh}$$

A7

0.40 pts

Calculate the values of K_a and ω_0 .

Answer:

$$K_a = 3.44 \cdot 10^4 \text{ s}^{-2}$$

$$\omega_0 = 2750 \text{ rad/s}$$

B1

1.00 pts

Express the intensity of light I incident on the photodiode in terms of δL , I_0 , λ , n , r and φ_0 . Here, φ_0 is the incident phase difference between the light passing through the fiber wound on the right and left supports, respectively, at $\delta L = 0$. Note: You don't have to follow the φ_0 sign. In other words, solutions differing by the change $\varphi_0 \rightarrow -\varphi_0$ are considered equivalent.

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After passing through the interferometer, waves with complex amplitudes

$$\frac{ri}{\sqrt{2}}E_0 e^{2ikn\delta L + i\varphi_0} \quad \text{and} \quad \frac{r}{\sqrt{2}}E_0 e^{-2ikn\delta L},$$

arrive at the splitter, so the total is

$$\begin{aligned} E &= \frac{ri}{2}E_0 e^{2ikn\delta L + i\varphi_0} - \frac{r}{2}E_0 e^{-2ikn\delta L} = -\frac{riE_0}{2} (e^{2ikn\delta L + i\varphi_0 + i\pi} + e^{-2ikn\delta L}) = \\ &= -\frac{riE_0}{2} (e^{2ikn\delta L + i\varphi_0/2 + i\pi/2} + e^{-2ikn\delta L - i\varphi_0/2 - i\pi/2}) e^{i\varphi_0/2 + i\pi/2} = -riE_0 \cos\left(\frac{4\pi}{\lambda}n\delta L + \frac{\varphi_0}{2} + \frac{\pi}{2}\right) \end{aligned}$$

$$I = EE^* = r^2 I_0 \sin^2\left(\frac{4\pi}{\lambda}n\delta L + \frac{\varphi_0}{2}\right)$$

Answer:

$$I = r^2 I_0 \sin^2\left(\frac{4\pi}{\lambda}n\delta L + \frac{\varphi_0}{2}\right)$$

B2

1.50 pts

Determine the numerical values of Y and f .

The oscillation period of the picture is $T = 63 \text{ ms} - 13 \text{ ms} = 50 \text{ ms}$, hence:

Half of the oscillation period corresponds to a change in $\ddot{y}$ from $-(2\pi f)^2 Y$ to $(2\pi f)^2 Y$.

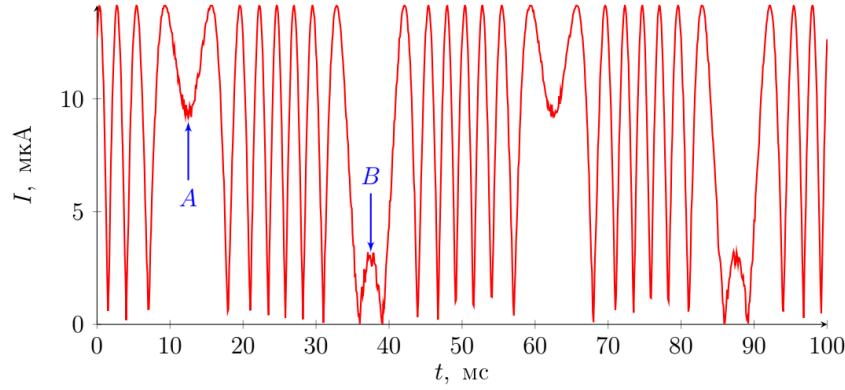
Let us assume that the argument x of the function $\sin^2(x)$, which describes the light intensity at point A , is in the interval $(0, \pi/2)$. Then at the point with practically zero current to the left of B it is equal to 7π . The amplitude of current fluctuations is $I_o = 14.2 \mu\text{A}$. Current values at points A and B : $I_A = 9.7 \mu\text{A}$, $I_B = 3.2 \mu\text{A}$. That means $x_B - x_A = 7\pi + \arcsin(\sqrt{I_B/I_o}) - \arcsin(\sqrt{I_A/I_o}) = 21.5$. As a

result,

$$x_B - x_A = \frac{4\pi}{\lambda}(\delta L_B - \delta L_A) = \frac{4\pi n}{\lambda} \cdot 2 \frac{(2\pi f)^2 Y}{K_a} \Rightarrow Y = (x_B - x_A) \frac{\lambda K_a}{32\pi^3 n f^2} = 1.46 \text{ } \mu\text{m}$$

Answer:

$$f = 20 \text{ Hz}$$



B3

0.30 pts

Using the theorem about the uniform distribution of energy over degrees of freedom, write down an expression for $\langle \delta L^2 \rangle$ —fluctuations δL at room temperature. Express your answer in terms of Boltzmann's constant k_B , T , E , D , N and d .

The described system has one degree of freedom, but the deformation energy at the two supports adds up, therefore

$$2\langle W \rangle = \frac{1}{2}k_B T \Rightarrow \langle \delta h^2 \rangle = k_B T \frac{8h^2}{E\pi^2 N D d^2}$$

Taking into account that $\langle \delta L^2 \rangle = \Lambda^2 \langle \delta h^2 \rangle$, we obtain

$$\langle \delta L^2 \rangle = 2k_B T \frac{ND}{Ed^2}$$

Answer:

$$\langle \delta L^2 \rangle = 2k_B T \frac{ND}{Ed^2}$$

B4

0.60 pts

If the temperature of this section of the optical fiber differs from room temperature by a small amount ΔT_i , then when light passes through it, phase $\varphi_i \neq \varphi_{i,0}$ occurs. Express $\Delta\varphi_i = \varphi_i - \varphi_{i,0}$ in terms of ΔT_i , ΔL_i , α , β , n and λ .

$$\Delta\varphi_i = \frac{2\pi}{\lambda}(n + \beta\Delta T_i)\Delta L_i(1 + \alpha\Delta T_i) - \frac{2\pi}{\lambda}n\Delta L = \frac{2\pi}{\lambda}(n\alpha + \beta)\Delta L_i\Delta T_i$$

Answer:

$$\Delta\varphi_i = \frac{2\pi}{\lambda}(n\alpha + \beta)\Delta L_i\Delta T_i$$

B5

0.70 pts

Express $\langle \Delta\Phi^2 \rangle$ in terms of $\langle \Delta T^2 \rangle$, ΔL , L , n , α , β and λ .

Let's calculate $\langle \Delta \Phi^2 \rangle$ directly:

$$\langle \Delta \Phi^2 \rangle = \left\langle \left(\sum \Delta \varphi_i \right)^2 \right\rangle = \left(\frac{2\pi}{\lambda} (n\alpha + \beta) \Delta L \right)^2 \left\langle \left(\sum \Delta T_i \right)^2 \right\rangle$$

Using the properties ΔT_i ,

$$\left\langle \left(\sum \Delta T_i \right)^2 \right\rangle = \left\langle \sum \Delta T_i^2 \right\rangle + \left\langle \sum \Delta T_i \Delta T_j \right\rangle = p \langle \Delta T^2 \rangle = \frac{L}{\Delta L} \langle \Delta T^2 \rangle.$$

Answer:

$$\langle \Delta \Phi^2 \rangle = \left(\frac{2\pi}{\lambda} (n\alpha + \beta) \right)^2 L \Delta L \langle \Delta T^2 \rangle$$

B6

1.00 pts

Express $\langle \Delta \Psi^2 \rangle$ in terms of $\langle \Delta \Phi^2 \rangle$.

The temperature fluctuations described earlier are a much slower process than the propagation of light, so for any $\Delta \Phi$ it holds that

$$\Delta \Psi = 2\Delta \Phi \quad \Rightarrow \quad \langle \Delta \Psi^2 \rangle = 4\langle \Delta \Phi^2 \rangle$$

Note that the answer differs by a factor of 2 from propagation along a fiber of length $2L$!

Answer:

$$\langle \Delta \Psi^2 \rangle = 4\langle \Delta \Phi^2 \rangle$$

B7

0.60 pts

Find the root-mean-square difference in the changes in the phase shift on the two arms of the interferometer $\langle (\Delta \Psi_1 - \Delta \Psi_2)^2 \rangle$. Express your answer using $\langle \Delta \Psi^2 \rangle$.

The phase shift on both interferometers is independent, therefore

$$\langle (\Delta \Psi_1 - \Delta \Psi_2)^2 \rangle = \langle \Delta \Psi_1^2 \rangle + \langle \Delta \Psi_2^2 \rangle = 2\langle \Delta \Psi^2 \rangle$$

Answer:

$$\langle (\Delta \Psi_1 - \Delta \Psi_2)^2 \rangle = 2\langle \Delta \Psi^2 \rangle$$

B8

1.00 pts

Consider that the sensor operates at $T = 305$ K and the magnitude of temperature fluctuations $\langle \Delta T^2 \rangle$ is $1.0 \cdot 10^{-6}$ K². What is the minimum acceleration $a_{\min}$ that can be measured using the proposed sensor?

Due to thermal fluctuations in the mechanical system, the length of the interferometer arms has a noise of

$$\sqrt{\langle \delta L^2 \rangle} = \sqrt{2k_B T \frac{ND}{Ed^2}} = 7.1 \cdot 10^{-12} \text{ m}$$

. Due to thermal fluctuations in the fiber, the phase difference has a noise of

$$\sqrt{\langle (\Delta \Psi_1 - \Delta \Psi_2)^2 \rangle} = \frac{4\pi\sqrt{2}}{\lambda} (n\alpha + \beta) \sqrt{\pi DN \Delta L} \cdot \sqrt{\langle \Delta T^2 \rangle} = 2 \cdot 10^{-3},$$

, which leads to noise in the calculated δL equal to $\frac{\lambda}{8\pi n} \cdot 2 \cdot 10^{-3} = 0.6 \cdot 10^{-10}$ m. Thus, the contribution from the second effect is an order of magnitude greater, and the minimum possible acceleration is:

Answer:

$$a_{\min} \approx K_a \cdot 0.6 \cdot 10^{-10} \text{ m} = 2 \cdot 10^{-6} \text{ m/s}^2$$